

Contact

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Education

Master of Artificial Intelligence & Machine Learning

University of Adelaide (Go8) | Australia
2024 – 2026 | GPA: 5.9/7

Bachelor of Engineering, Computer Science

Anna University (Kumaraguru College of Technology) | India
2020 – 2024 | GPA: 8.08/10

Skills

- LLM Fine-Tuning (LoRA/QLoRA, Unsloth)
- LLM Agent Orchestration & Guardrails
- RAG, Vector Search & Embeddings
- Natural Language Processing (NLP)
- Computer Vision / CNNs
- Application Security (DAST)
- Full-Stack Development (MERN)
- Authentication & Row-Level Security
- Model Evaluation & Rubric Judging
- Data Processing & Visualization
- Access Control
- Data Labeling and Locking
- Spotting abnormal Behavior

CHRISTEN LOYOLA

Software Engineer
AI & Machine Learning Specialist

Professional Experience

2025
–
2026

Software Engineer

Ahfy | Adelaide, Australia | Sep 2025 – Aug 2026

Client Portal: Designed and built, from the ground up, the B2B client portal that university and school leadership (chancellors, principals, senior staff) use to review student data, owning it end to end from initial design to production.

Shared Dev Database: Created and maintained a standardized development database for the team (table structures, naming conventions, security policies) and managed environment configuration and secrets.

Authentication: Built Supabase authentication with SQL-based role detection and access control, so users with higher clearance can reach sensitive information.

Row-Level Security: Designed and enforced RLS policies for multi-role users, with conditional data visibility based on login context.

Dashboard Features: Developed features end to end in React and TypeScript, from filtering to a bulk CSV import flow with deduplication, validation and friendly error handling, backed by a table-based queuing system built in Supabase with custom backend processing functions.

Support Tooling: Built a support and bug-report flow with image attachments and a sessionStorage breadcrumb buffer that automatically attaches reproduction context to each report.

Edge-Function Security: Secured a Supabase edge function with Bearer-JWT authentication, Postgres role-based access control and magic-byte file validation, behind a structured error-code contract for clean client handling.

UI Ownership: Owned the dashboard’s UI end to end in React and TypeScript: reusable, accessible components including tables, pagination and filtering, loading, empty and null states, and render-bug fixes such as tables flashing empty on filter changes.

Debugging & Hardening: Hardened the dashboard across many fixes, including a one-week stretch of 14 pushes spanning bug fixes and new features, and a production outage traced to an environment-variable naming mismatch.

Code Review: Reviewed pull requests as part of the team workflow, catching a fail-open authorization path in a security-sensitive access-control feature before it shipped.

2023

Internship Trainee Developer

Power Links Consortium | Jun 2023 – Jul 2023

- Built 3D models and the tooling on top of them on the iTwin platform, developed primarily in React; delivered the 3D visualization tooling for the company’s industrial clients.
- Built a configuration log reader that parses log files, including troubleshooting for the configuration log.

2023

Online Intern

Campalin Innovations | Jan 2023 – Feb 2023

- Built a wine-quality ranking classifier grading wines as good or poor from seven engineered attributes, trained on a labelled wine dataset.

Language

- English
- Tamil
- Hindi
- Arabic

Technical Competencies

- Python / TypeScript / JavaScript
- PyTorch / TensorFlow / Keras
- React.js / Next.js / Vue (Nuxt 3)
- Node.js / Express / MongoDB
- FastAPI / Django / REST APIs
- SQL / PostgreSQL / Supabase
- AWS (S3, ECS Fargate, ECR, CloudWatch)
- Docker / Redis (Upstash)
- OpenAI Agents SDK / OpenAI & Gemini APIs
- Modal / vLLM / Vercel
- OWASP ZAP / pytest / Pydantic
- Lovable / FlutterFlow
- Claude Code / Cursor / Codex
- Microsoft Sentinel

Links

LinkedIn

[linkedin.com/in/christen-loyola-912858221](https://www.linkedin.com/in/christen-loyola-912858221)

GitHub

github.com/Christen1609

Portfolio

christenloyolaportofliowebsite.vercel.app

Projects

2026

Pensure: AI-Agent Vulnerability Scanner

Prototype · Python / FastAPI / OpenAI Agents SDK / AWS

- Built an end-to-end AI security-scanning pipeline: OWASP ZAP scans a deliberately vulnerable target (OWASP Juice Shop), an LLM judge agent (OpenAI Agents SDK + Gemini 2.5 Flash) triages each Pydantic-normalized finding with a confidence score, false-positive flag and written remediation, and versioned JSON reports land in AWS S3.
- Engineered a bounded, tool-using agent loop with deterministic guardrails: the judge re-verifies findings against the live target, and a reconciliation step drops hallucinated findings, restores scanner-immutable fields and fails closed on malformed output.
- Shipped an async job API (202 + polled job id, live progress) and a Nuxt 3 + TypeScript dashboard; deployed as one multi-stage Docker image on AWS ECS Fargate (ECR, least-privilege IAM, CloudWatch-to-SNS alerting), with a dependency-injected pytest suite testing the pipeline without live services.

2026

Radiant Mind: AI Learning Web App

Mar 2026 – Jun 2026

- Built an AI-powered learning web app on Next.js, React, TypeScript, Supabase and Vercel, with a 3-layer AI content pipeline that turns any topic into a coherent narrative arc, grounded in Wikipedia to reduce hallucination.
- Engineered an automated rubric judge scoring generated content across 8 quality properties (pass threshold $\geq 12/16$), reused to gate production output, filter training data and evaluate models.
- Fine-tuned Qwen2.5-7B with Unsloth (LoRA/QLoRA) on a ~1,146-row judge-labeled dataset; picked the best configuration via a hyperparameter sweep and a blind, judge-scored bake-off; served on Modal (vLLM, scale-to-zero L4 GPU, OpenAI-compatible endpoint).
- Hardened for production with a JWT auth gate, Supabase RLS, tiered Upstash Redis rate limiting, and multi-layer caching cutting topic load time from 2.2s to 0.75s.

2026

Two-Stage RAG Engine (oolee.com.au clone)

Apr 2026

- Built a two-step embedding engine powering retrieval-augmented generation over a vector database, with a personalization layer tailoring retrieved results to the user; Next.js + TypeScript, Gemini embeddings, Gemini Flash generation.

2025

Web App for Regex Pattern Matching & Replacement

Aug 2025 – Sep 2025

- Full-stack Django + React app using OpenAI's LLM for text pattern recognition and data cleaning: users upload CSV/Excel, describe patterns in plain language, and get generated regex via REST API, with match highlighting, column replacements, preview/download and regex explanation.

2023

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2024

Underwater Marine Animal Identification System

- **Team Lead.** Identified the type and name of marine creatures from features extracted across 7,000 images using VGG16, with classification via a custom Convolutional Neural Network (CNN).

More projects: github.com/Christen1609

Leadership & Activities

2023

Marketing Lead – Bharat Blockchain Yatra (Nov 2023): Led the marketing team and coordinated strategy between the organizers and the host university; ran outreach to Heads of Department and principals and managed social-media promotion.